

Our project compares three treatments from a small simulated trial and asks which one to use. Everything we show comes from one script, and every probability is computed exactly.

B has the highest raw success rate, 70 out of 100, but with only 100 patients each the gaps of 4 and 8 percentage points could be noise. The question is how much the data really tell us.

The likelihood depends on the data only through the number of successes, so the table is all we need. A uniform prior expresses no preference, and the posterior is again Beta: add successes to the first parameter and failures to the second.

The three posteriors overlap a lot. B is ahead with posterior mean 0.6961, but its credible interval overlaps both others. The posterior mean is the Bayes estimator under squared-error loss; median and mode are in the report.

The shaded area is the probability that the first treatment is better. B beats A with probability 0.882, but a gain of more than 5 points is less certain, 0.667. The p-value of 0.116 answers a different question.

Because the parameters are whole numbers, the densities and CDFs are polynomials, so these integrals have exact rational values. We checked them with a second exact formula, numerical integration and simulation.

B is the most likely best treatment, 0.681, then C, 0.248. On the right is the expected loss of each choice, the success rate we would expect to give up by not picking the true best.

Both natural loss functions pick B, and the choice does not change under any of four priors, including a fairly strong Beta(10,10) prior centred on 50 percent.

B is clearly better than A, but only modestly ahead of C. Even if the observed rates held exactly, we would need roughly 750 patients per arm to be 95 percent sure B is best.

Our problem has exact answers, so it is a clean test of the Module 5 approximations. After reproducing the course example, both methods are accurate to about one part in ten thousand, and Lindley's terms add up exactly for the odds.

Our recommendation is B. It has the highest estimated success rate, the highest chance of being best and the smallest expected loss, under every prior. B versus C is the open question.